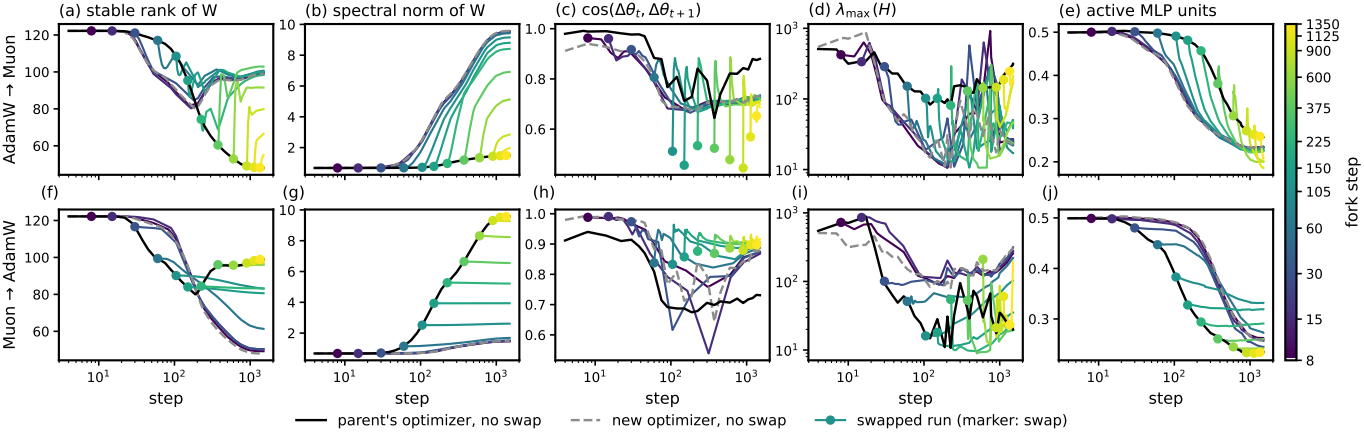

(a) stable rank of W
(b) spectral norm of W
(c) $\cos(\Delta\theta_t, \Delta\theta_{t+1})$
(d) $\lambda_{\max}(H)$
(e) active MLP units
(f)
(g)
(h)
(i)
(j)
AdamW → Muon
Muon → AdamW
step
fork step
1350
1125
900
600
375
225
150
105
60
30
15
8
parent's optimizer, no swap
new optimizer, no swap
swapped run (marker: swap)